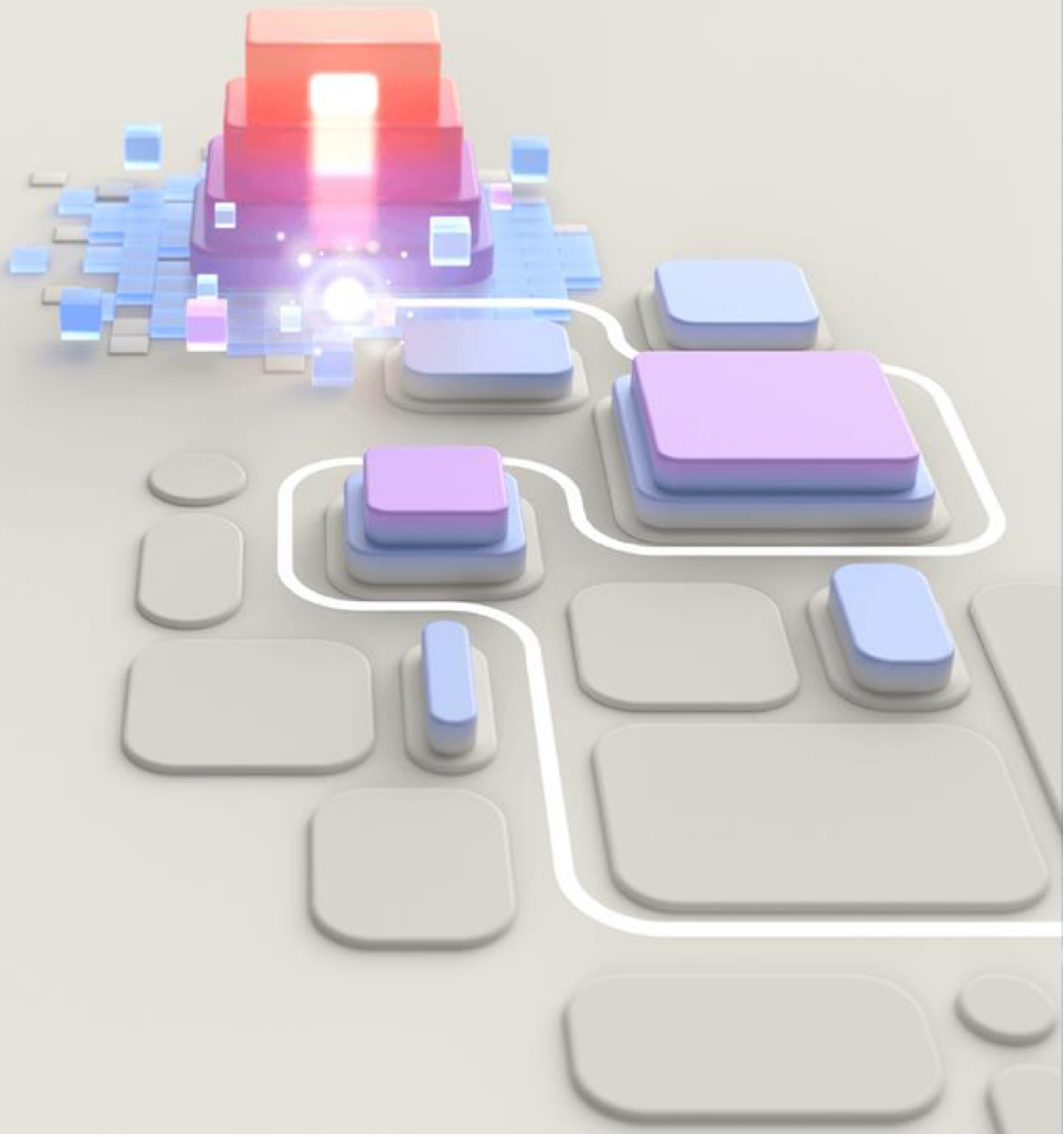
Microsoft

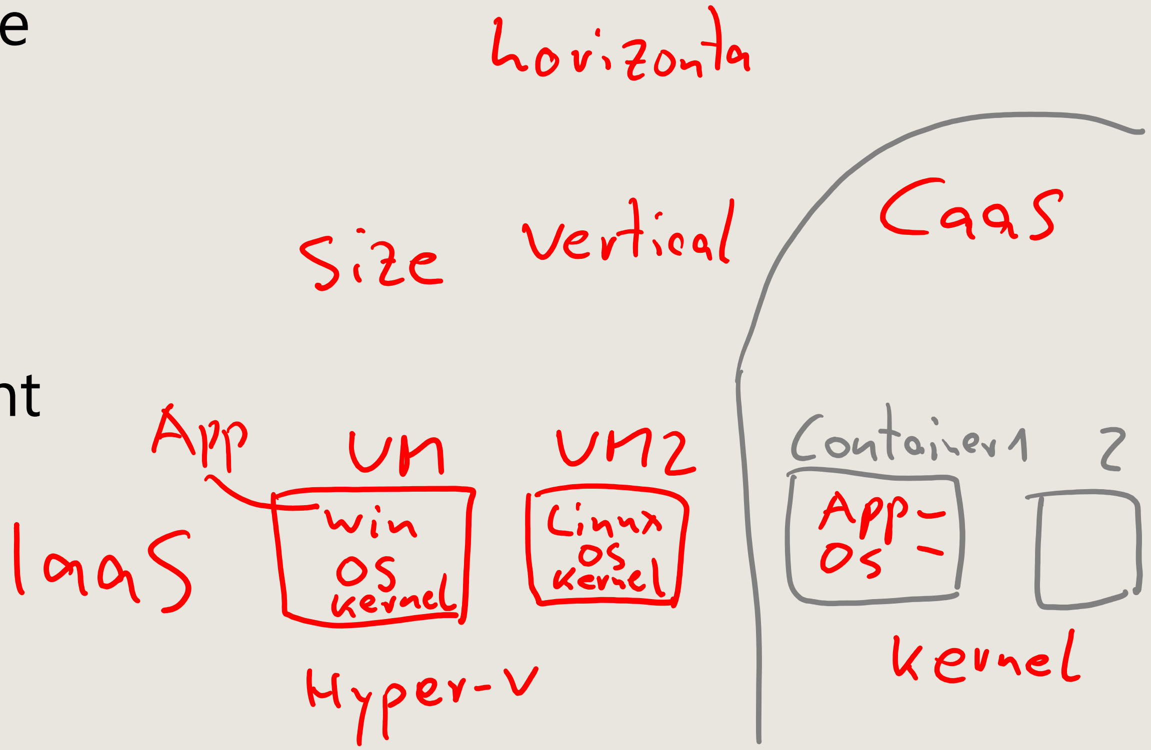


AZ-104

Administer
Azure Virtual Machines

AZ-104 Agenda

01: Administer Identity
02: Administer Governance and Compliance
03: Administer Azure Resources
04: Administer Virtual Networking
05: Administer Intersite Connectivity
06: Administer Network Traffic Management
07: Administer Azure Storage
08: Administer Azure Virtual Machines
09: Administer PaaS Compute Options
10: Administer Data Protection
11: Administer Monitoring



Learning Objectives - Administer Azure Virtual Machines

- Introduction to Azure Virtual Machines
- Configure Virtual Machine Availability
- Lab 08 – Manage Virtual Machines

Introduction to Azure Virtual Machines

Learning Objectives – Introduction to Azure Virtual Machines

- Review Cloud Services Responsibilities
- Plan Virtual Machines
- Determine Virtual Machine Sizing
- Determine Virtual Machine Storage
- Connect to Virtual Machines
- Demonstration - Create and connect to virtual machines
- Learning Recap

Exam objectives

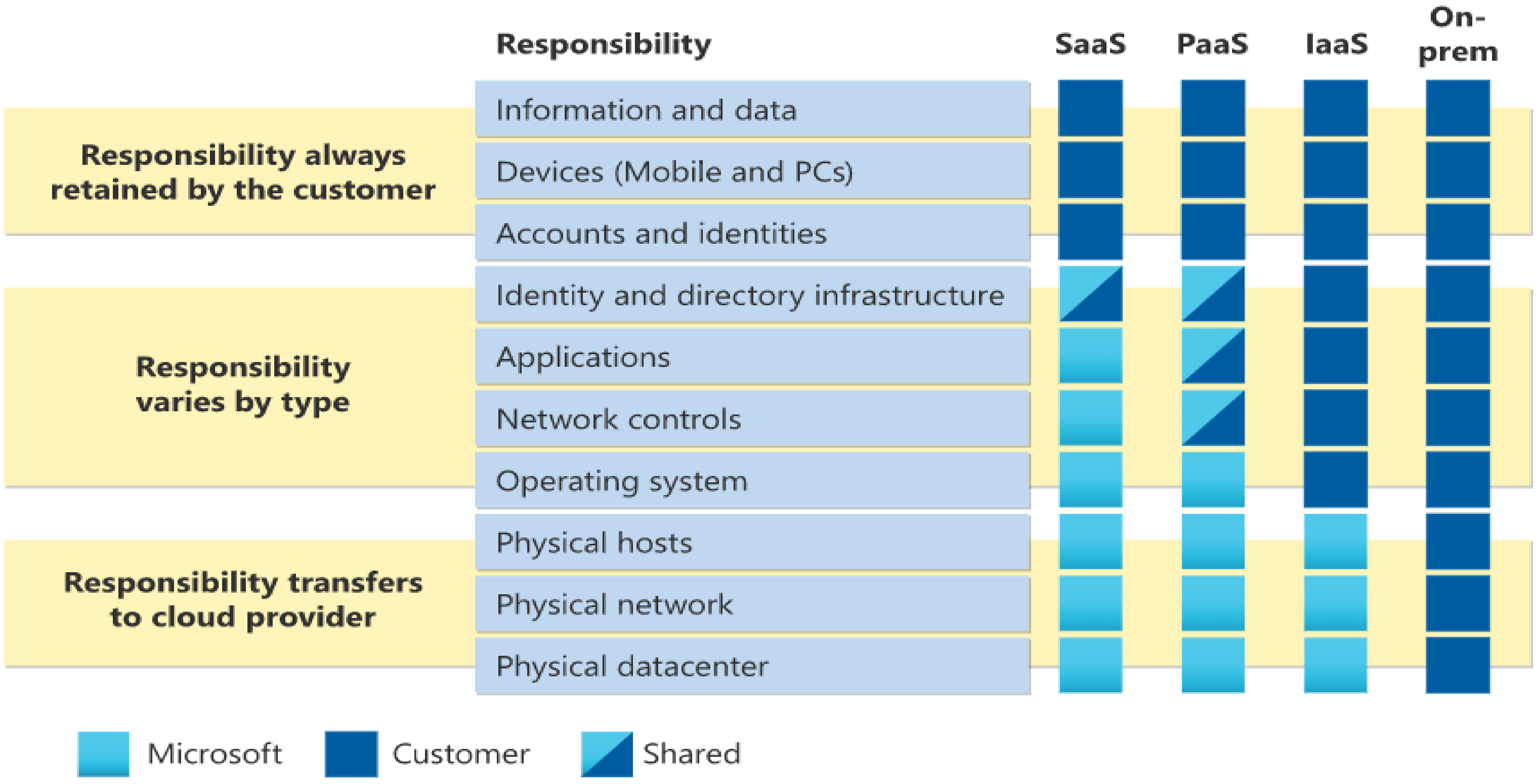
Create and configure virtual machines (VMs)

- Create a VM
- Move VMs between resource groups
- Manage VM sizes
- Manage VM disks

Configure secure access to virtual networks

- Implement Azure Bastion

Review Cloud Services Responsibilities



Plan Virtual Machines

* Same Region

- Start with the network
- Name the virtual machine
- Choose a location
  - Each region has different hardware and service capabilities
  - Locate Virtual Machines as close as possible to your users and to ensure compliance and legal obligations
- Consider pricing

Determine Virtual Machine Sizing

Type	Description
General purpose	Balanced CPU-to-memory ratio.
Compute optimized	High CPU-to-memory ratio.
Memory optimized	High memory-to-CPU ratio.
Storage optimized	High disk throughput and I/O.
GPU	Specialized virtual machines targeted for heavy graphic rendering and video editing.
High performance compute	Our fastest and most powerful CPU virtual machines

Determine Virtual Machine Storage

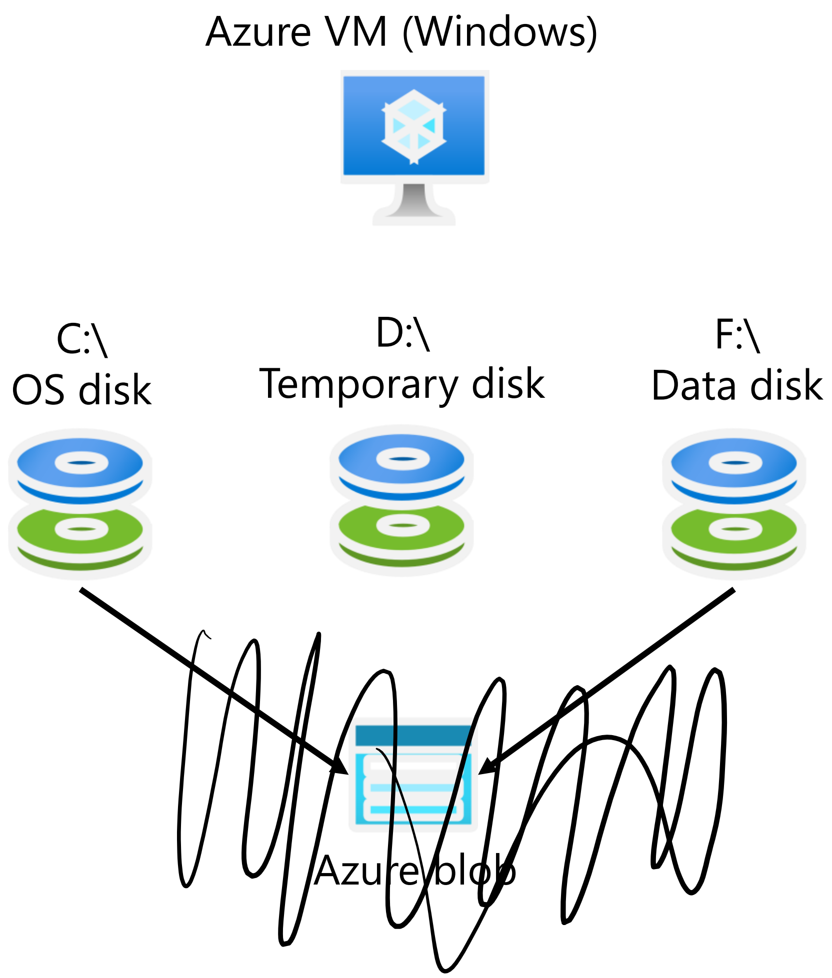
Each Azure VM has two or more disks:

- OS disk
- Temporary disk (not all SKUs have one, content can be lost)
- Data disks (optional)

OS and data disks reside in Azure Storage accounts:

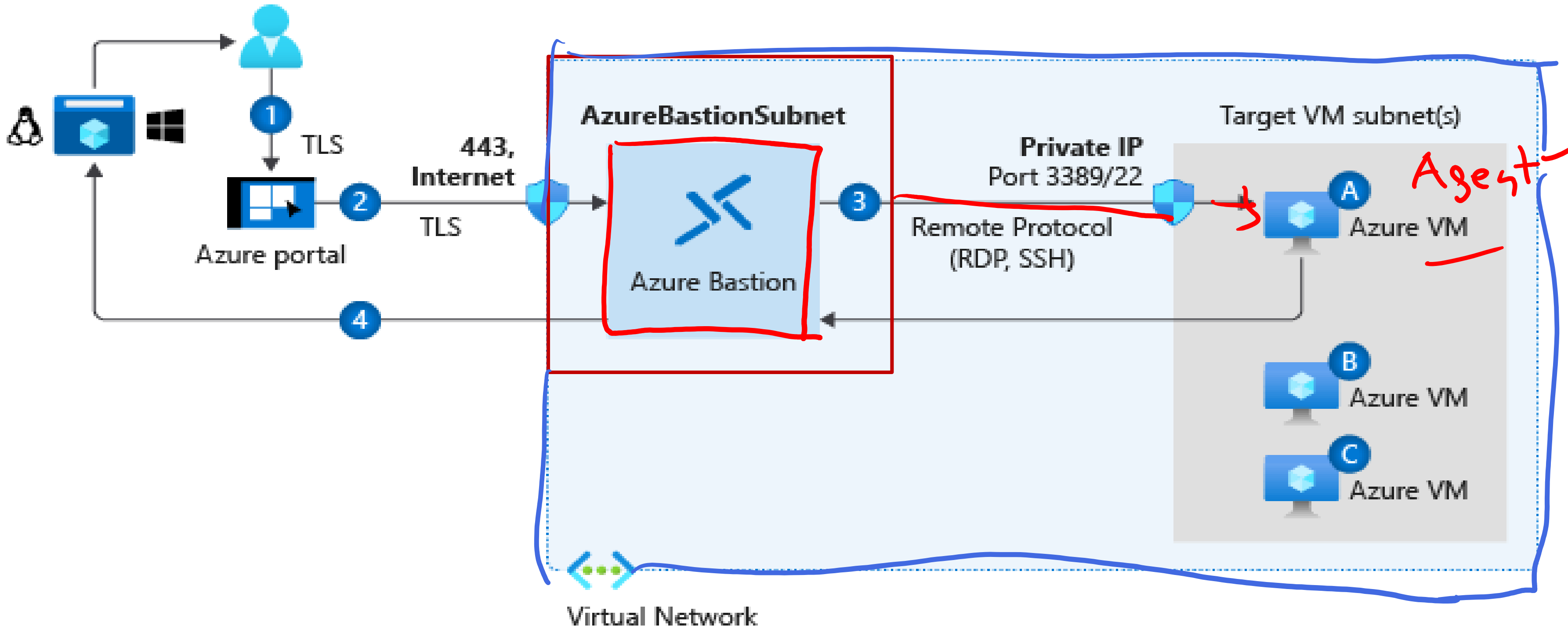
- Azure-based storage service
- Standard (HDD, SSD) or Premium (SSD), or Ultra (SSD)

Azure VMs use managed disks



Connect to Virtual Machines

Portal
Run Command



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

Configure Virtual Machine Availability

Configure Azure Virtual Machine Availability Introduction

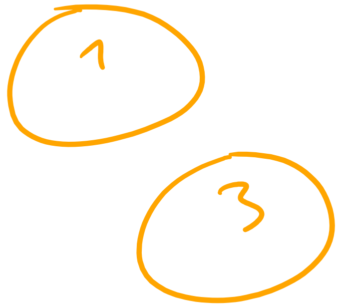
- Plan for Maintenance and Downtime
- Setup Availability Sets
- Review Update and Fault Domains
- Review Availability Zones
- Compare Vertical to Horizontal Scaling
- Create and Configure Scaling
- Demonstration – Virtual Machine Scaling
- Bring Azure innovation to your hybrid environments with Azure Arc (optional)
- Learning Recap

Exam objectives

Create and configure virtual machines

- Deploy virtual machines to availability zones and availability sets
- Deploy and configure an Azure Virtual Machine Scale Sets

Setup Availability Sets



Avail Zone
= Data Center

Instance details

Name * ⓘ avset01

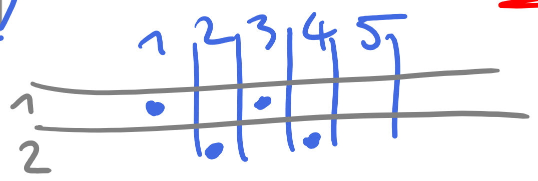
Region * ⓘ (US) East US

Fault domains ⓘ Rack 2

Update domains ⓘ 5

Use managed disks ⓘ No (Classic) Yes (Aligned)

Two or more instances in Availability Sets = 99.95% SLA



Configure multiple Virtual Machines in an Availability Set

Configure each application tier into separate Availability Sets

Combine a Load Balancer with Availability Sets

Use managed disks with the Virtual Machines

Review Availability Zones

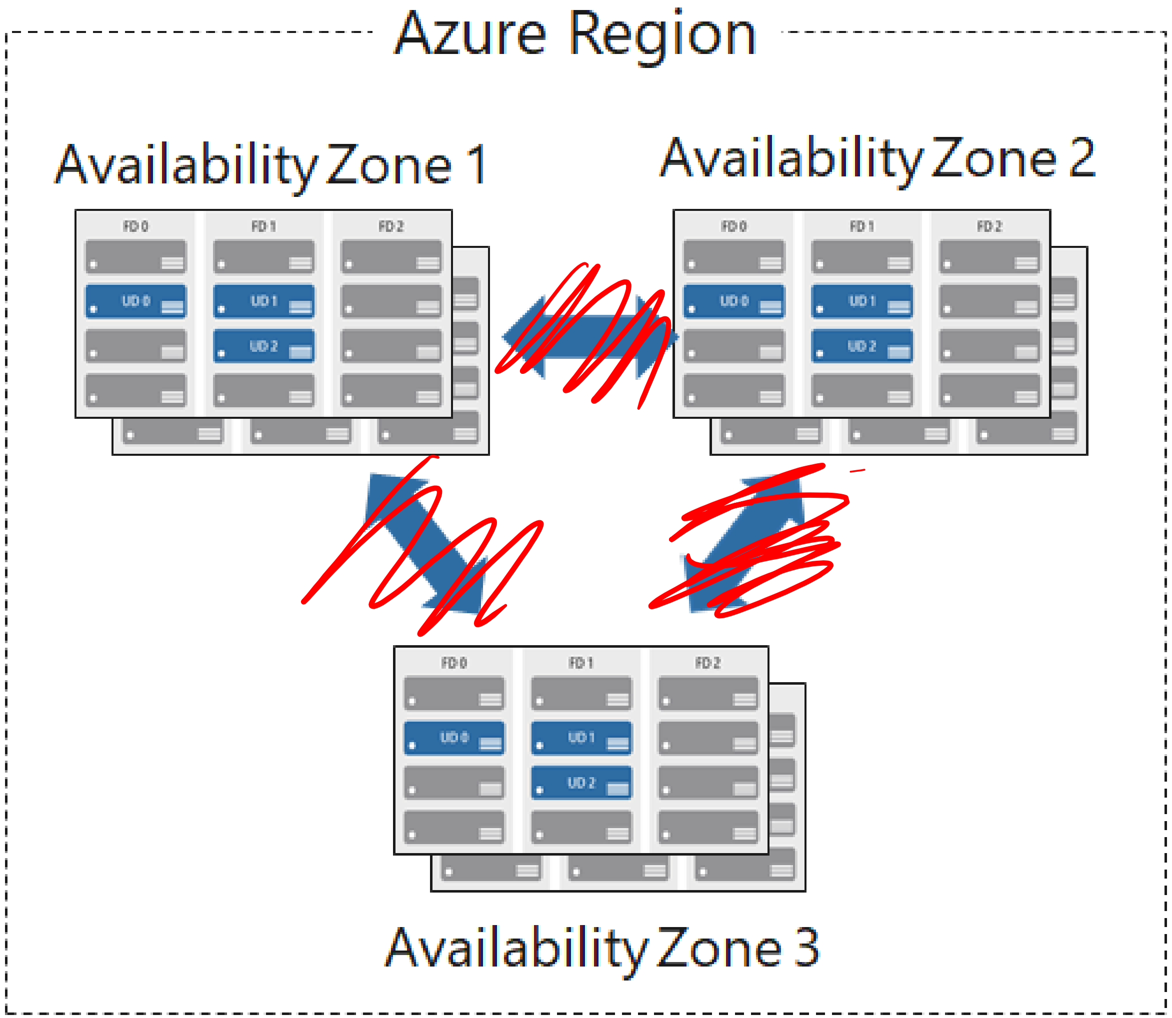
Unique physical locations in a region

Includes datacenters with independent power, cooling, and networking

Protects from datacenter failures

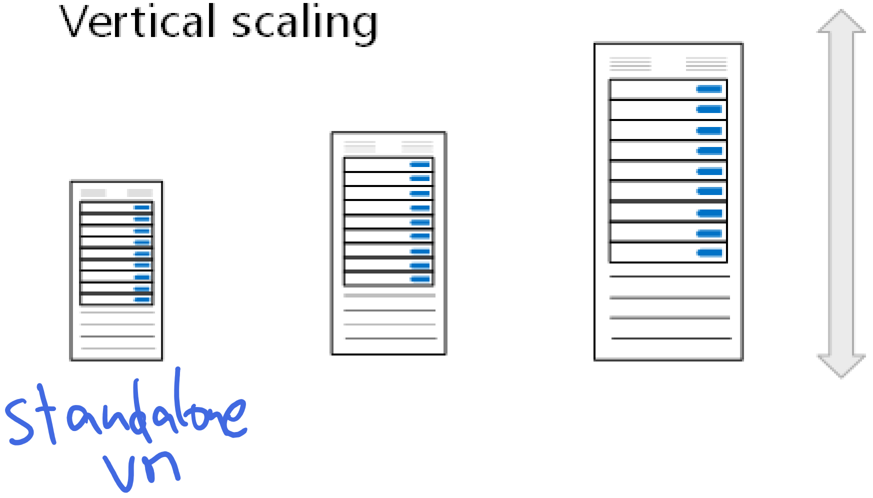
Combines update and fault domains

Provides 99.99% SLA

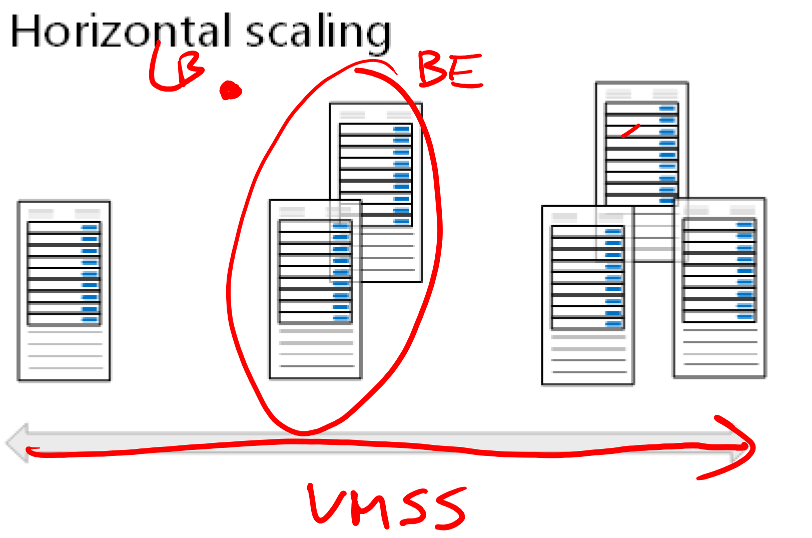


VMSS — uniform
flex

Compare Vertical to Horizontal Scaling



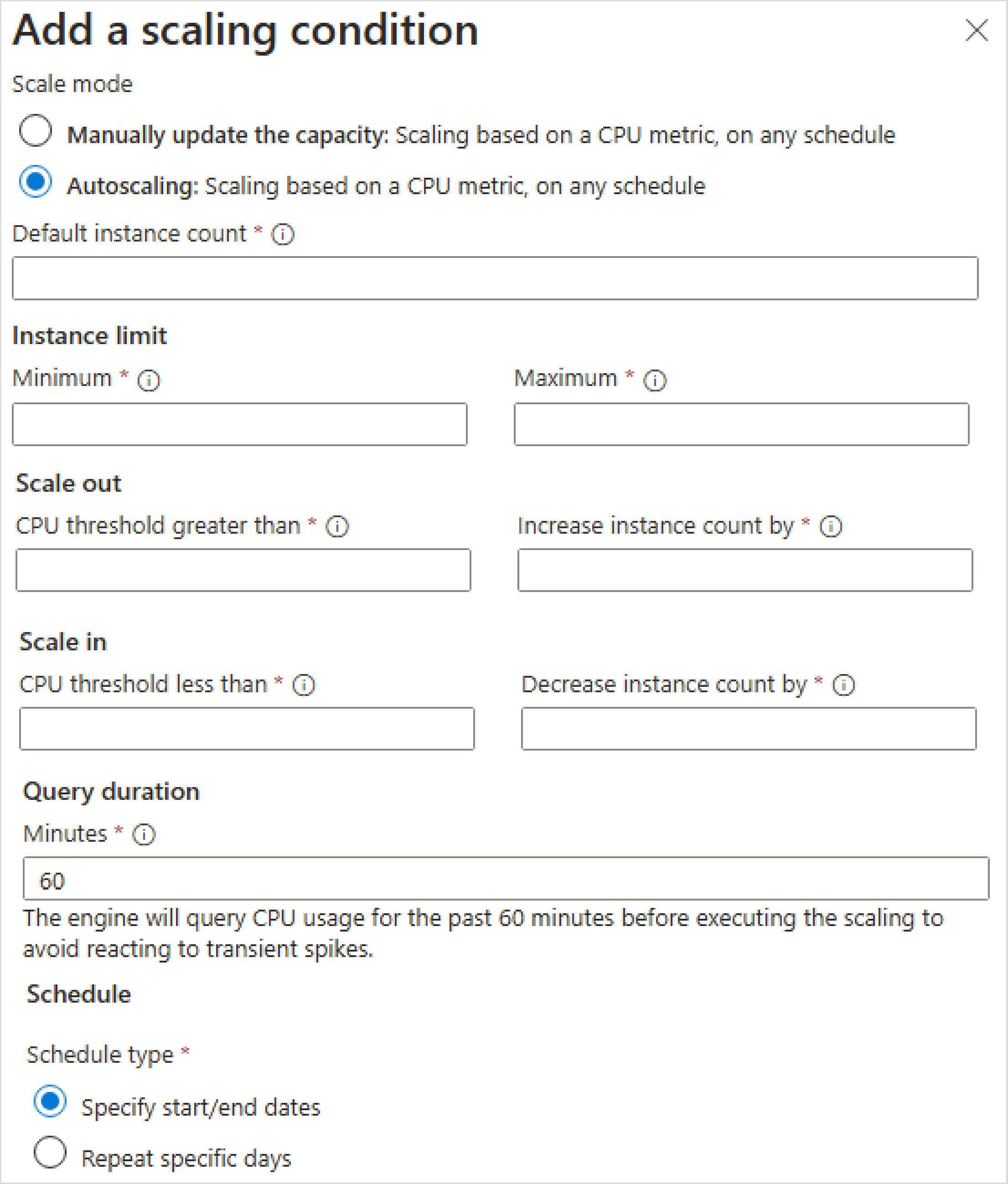
Vertical scaling (scale up and scale down) is the process of increasing or decreasing power to a single instance of a workload; usually manual



Horizontal scaling (scale out and scale in) is the process of increasing or decreasing the number of instances of a workload; frequently automated

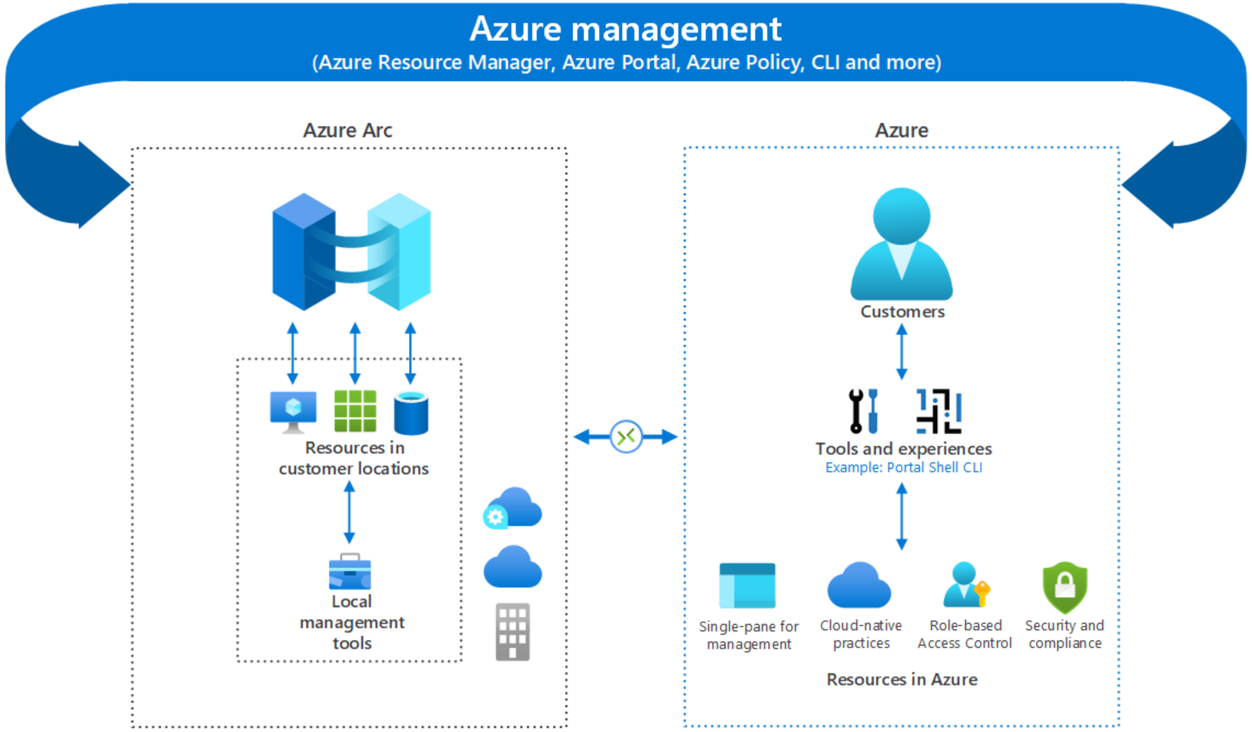
Create and configure scaling

- Manually update or autoscale
- Define a minimum, maximum, and default number of VM instances
- Create more advanced scale sets with scale out and scale in parameters



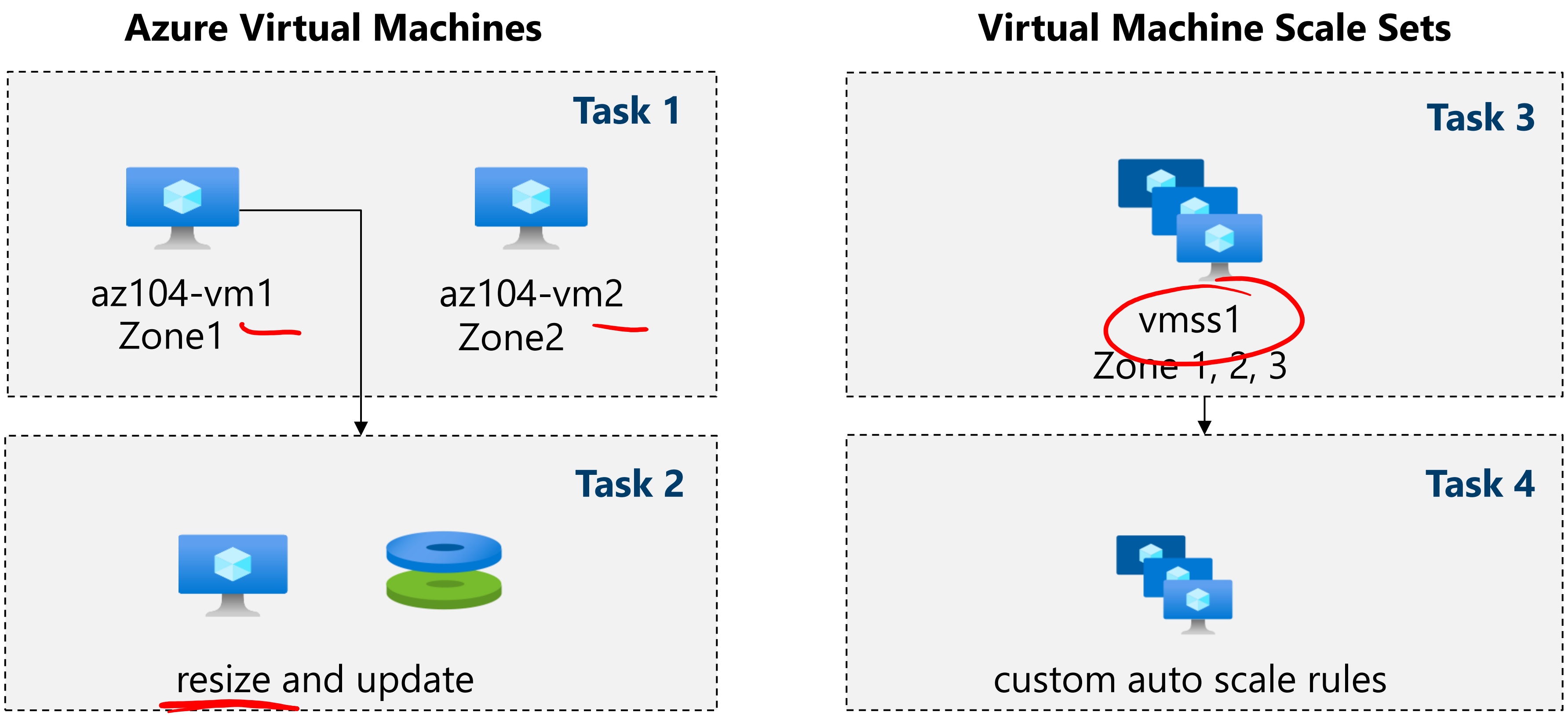
Bring Azure innovation to your hybrid environments with Azure Arc

- Manage your entire environment of existing non-Azure and/or on-premises resources
- Manage virtual machines, Kubernetes clusters, and databases as if they are running in Azure.
- Use familiar Azure services and management capabilities, regardless of where your resources live
- Use traditional IT Ops while introducing DevOps practices to support new cloud native patterns in your environment



Lab – Manage Virtual Machines

Lab 08 – Architecture diagram



Task 5: Create a virtual machine using Azure PowerShell (option 1)
Task 6: Create a virtual machine using the CLI (option 2)

End of presentation